Project Briefing: Enterprise GenAI RAG SaaS Chatbot

1. Project Overview

"I developed a **Retrieval-Augmented Generation (RAG) based SaaS chatbot** designed for enterprise knowledge management.
The system allows users to query internal documents and receive **accurate, context-aware responses** using a combination of vector search and large language models."

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2. Problem Statement

"Organizations often struggle with:

- Scattered documentation
- Inefficient knowledge retrieval
- Hallucinations from standalone LLMs

So I designed a system that:

- Grounds responses in **trusted internal data**
- Reduces hallucinations
- Enables **secure, multi-user access**"

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3. System Architecture

"Architecturally, the system follows a **RAG pipeline**:

1. **Data Ingestion Layer**
2. **Vector Database (FAISS)**
3. **Retriever**
4. **LLM Generator**
5. **API Layer (FastAPI)**
6. **Authentication Layer (JWT)**

This ensures modularity, scalability, and production readiness."

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4. Data Ingestion Pipeline

"I implemented a document ingestion pipeline that:

- Loads enterprise documents using `UnstructuredFileLoader`
- Splits them into chunks using `RecursiveTextSplitter`
- Converts them into embeddings using **Hugging Face sentence-transformers**
- Stores them in a **FAISS vector database**

This enables efficient semantic search."

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5. Vector Search (FAISS)

"I used **FAISS** for:

- High-performance similarity search
- Low-latency retrieval
- Local deployment without external dependencies

Each user query is converted into an embedding and matched against stored document chunks."

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6. LLM Integration (Hugging Face)

"I integrated a **free open-source LLM (Flan-T5)** using Hugging Face to:

- Generate answers based only on retrieved context
- Avoid hallucinations via prompt engineering

The prompt strictly enforces:

> 'Only answer from the provided context, otherwise say you don't know.'"

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7. Backend (FastAPI)

"I built the backend using **FastAPI**, which provides:

- High performance (async-ready)
- Clean API design
- Easy integration with frontend and services

Main endpoints:

- `/signup` → user registration
- `/login` → JWT authentication
- `/ask` → protected chatbot endpoint”

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8. Authentication & Security

“I implemented **JWT-based authentication**:

- Users must log in to access the chatbot
- Tokens are validated for each request
- Passwords are securely hashed using bcrypt

This makes the system suitable for **multi-tenant SaaS environments**.”

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9. Key Features

- Retrieval-Augmented Generation (RAG)
- Context-aware responses
- Open-source LLM (no API cost)
- Secure authentication (JWT)
- Scalable API architecture
- Modular design for future upgrades

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10. Challenges & Solutions

This part is VERY important for interviews

Challenge 1: LangChain breaking changes

- Solution: Migrated to new module structure (`langchain_huggingface`, etc.)

Challenge 2: Model compatibility issues

- Solution: Adjusted Hugging Face pipeline to match supported tasks

Challenge 3: FAISS security restrictions

- Solution: Enabled safe deserialization with controlled environment

Challenge 4: Running without OpenAI API

- Solution: Switched to Hugging Face open-source models

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11. Real-World Use Cases

"This system can be used for:

- Enterprise knowledge assistants
- Customer support automation
- Internal IT/helpdesk bots
- Banking & compliance document search
- Legal and policy Q&A systems"

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12. Future Enhancements

"To make it production-grade SaaS, I would:

- Add Redis caching for performance
- Implement multi-tenant data isolation
- Use scalable vector DBs like Pinecone or Weaviate
- Deploy on Kubernetes
- Add monitoring (Prometheus + Grafana)
- Integrate streaming responses"

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13. Why This Project Matters

"This project demonstrates:

- End-to-end GenAI system design
- Real-world LLM application (not just notebooks)
- Backend engineering + AI integration
- Production-level thinking (security, scalability, cost optimization)"

Pro Interview Tip (VERY IMPORTANT)

If they ask:

> “*What makes your project strong?*”

Say:

> “It’s not just a chatbot — it’s a **production-ready GenAI system** with retrieval grounding, authentication, and scalable architecture.”
